

MAT 504

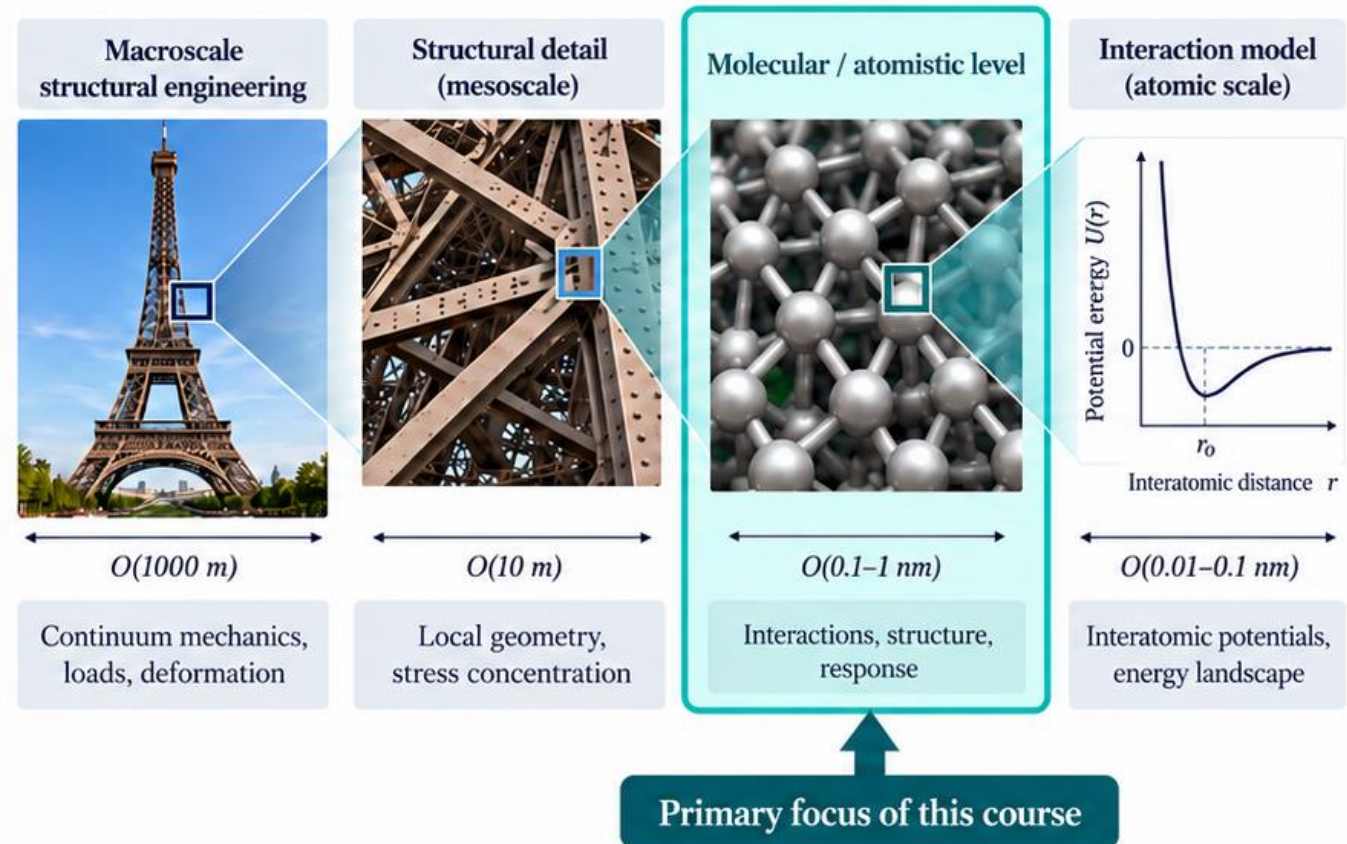
Computer Modelling in Materials Science

Week 1 — Introduction to Molecular Modeling
and Simulation

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Sabanci University

Model hierarchy: the Eiffel Tower example

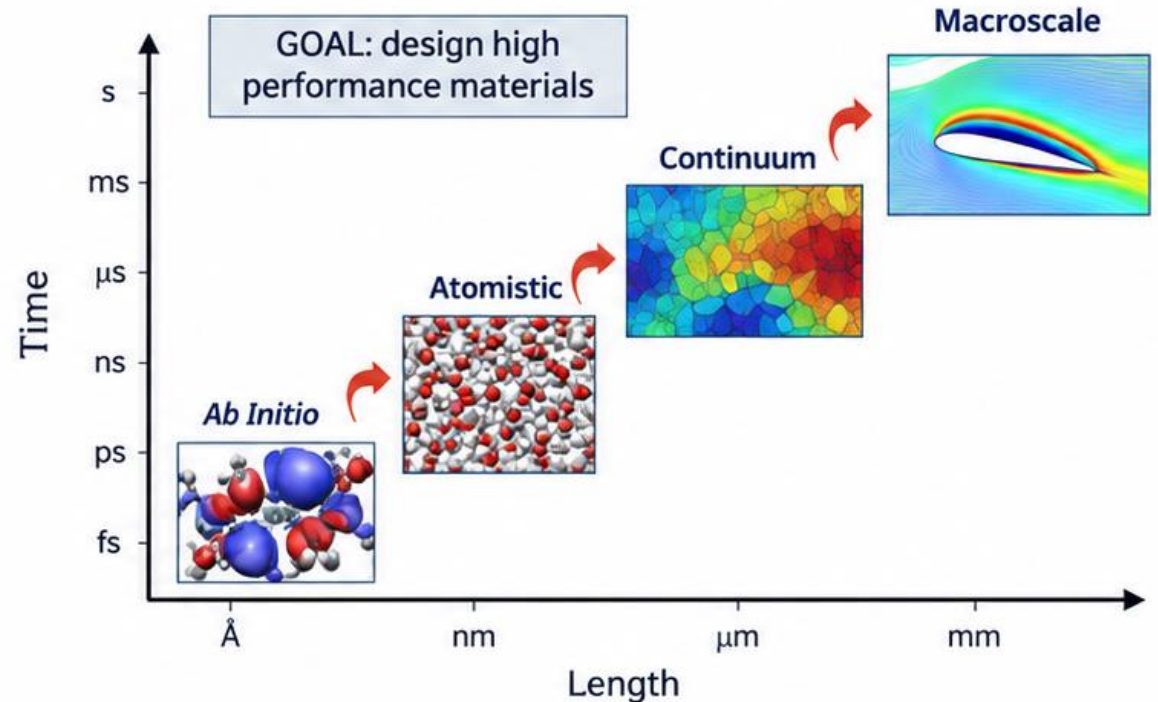
Engineering questions can be studied at different levels of description. **Our focus is the molecular level.**



What this course is about

Molecular simulation asks how microscopic interactions generate macroscopic behavior.

- We will focus on classical molecular dynamics, with Monte Carlo as a sampling foundation.
- We will treat simulation as a scientific workflow: model choice, setup, analysis, uncertainty, and interpretation.
- Advanced weeks will cover free-energy methods, rare-event kinetics, and machine learning for molecular simulation.



Length and time scales connect atomistic information to engineering models.

Working principle: the method should follow the scientific question, not the other way around.

Course Assessment

50%

Homework assignments

Computational exercises, analysis, interpretation, and possible oral discussion.

40%

Individual project

- Proposal 10%
 - Final report 25%
 - Presentation 15%
-

10%

Attendance and participation

Lecture Structure

Each week combines a lecture with computational practice.

2 h lecture + 1 h hands-on session

Homework is discussed after a delay so that late submissions remain possible before solutions are shown.



Assignments are solved in class approximately two weeks after they are assigned.

Course Outline

1	Why molecular simulation?	Models, scales, and an engineering example
2	Sampling foundations	Statistical mechanics and Monte Carlo
3-6	Core molecular dynamics	Integration, force fields, workflow, analysis, convergence
7-11	Advanced sampling and rare events	Umbrella sampling, REMD, metadynamics, TIS, FFS
12-13	Machine learning	Descriptors, supervised learning, trajectory analysis, CVs
14	Integration	Choosing methods for scientific questions

Important AI Use Policy

Use tools. Understand your work.

- Documentation, online resources, and generative AI may be used as support.
- Submitted code, figures, and interpretations remain your responsibility.
- Homework and project grading includes individual discussion of your submitted work.
- You may be asked to explain code line by line, reproduce an analysis step, justify a technical choice, or make a small modification to your code.

Homework assignments 45% of final course grade

Computational exercises, analysis, interpretation, and individual oral discussion of submitted work.

30%

Submitted work

Correctness and completeness of code, figures, analysis, written interpretation, and reproducibility of the submitted notebook.

70%

Oral assessment

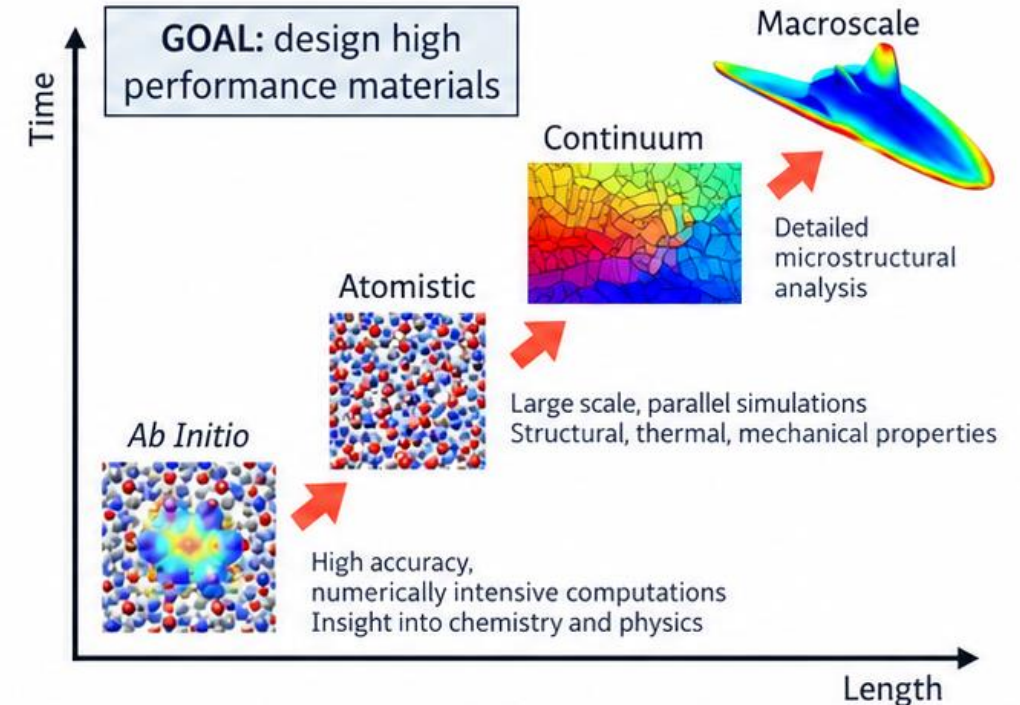
Individual discussion of your submitted work: understanding of code, methods, results, and interpretations.

If it is in your submission, you should be able to explain it.

Opening Question

How do we “run an experiment” on a material or molecule before we can fully test it in the lab?

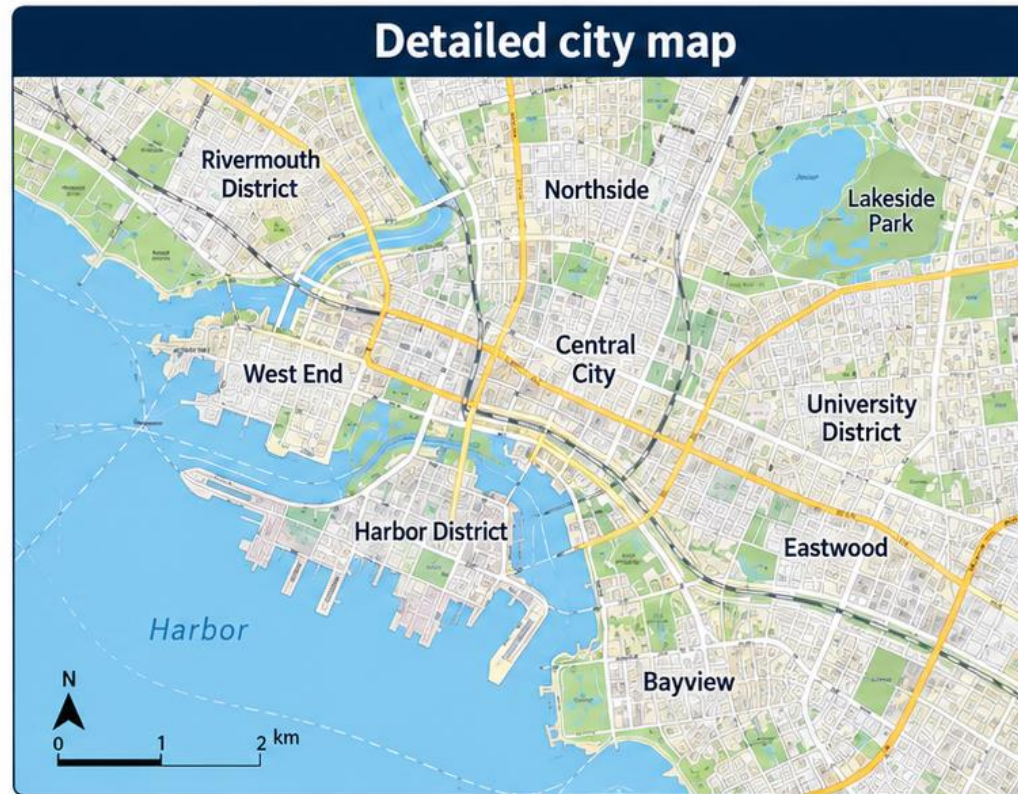
- Choose an appropriate level of description.
- Build a model that retains the physics needed for the question.
- Use computation to generate data from that model.
- Interpret the result with awareness of assumptions and limitations.



This is the recurring logic of the course.

What is a Model?

A model keeps the structure relevant to the question and discards unnecessary detail.



High detail, but not always useful for every question.

preserves
relevant structure



discards
unnecessary detail

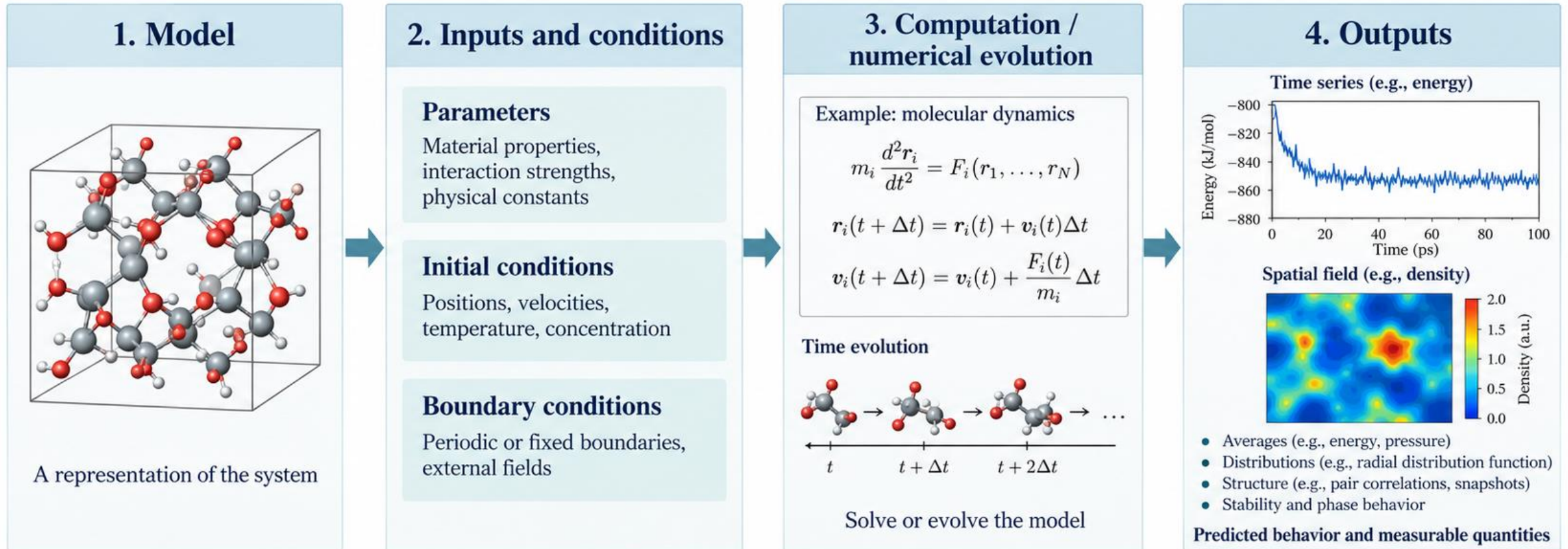


Simplified, but useful when the goal is routes and connectivity.

A useful model is not the most detailed one; it is the one that keeps the right details.

What is a Simulation?

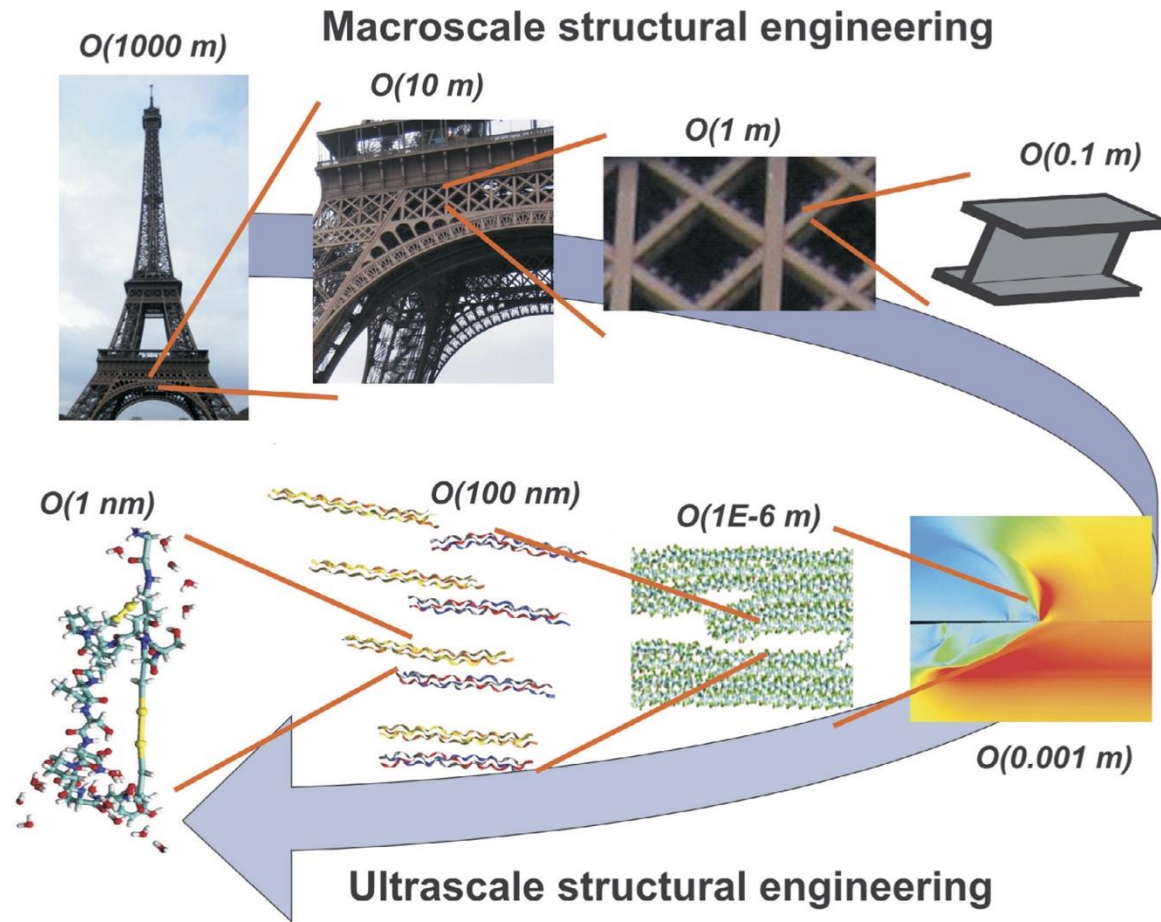
A simulation is the procedure of using a model to predict behavior under specified conditions.



Model + conditions + numerical procedure → prediction

Model = representation. Simulation = running that representation to ask: what happens if...?

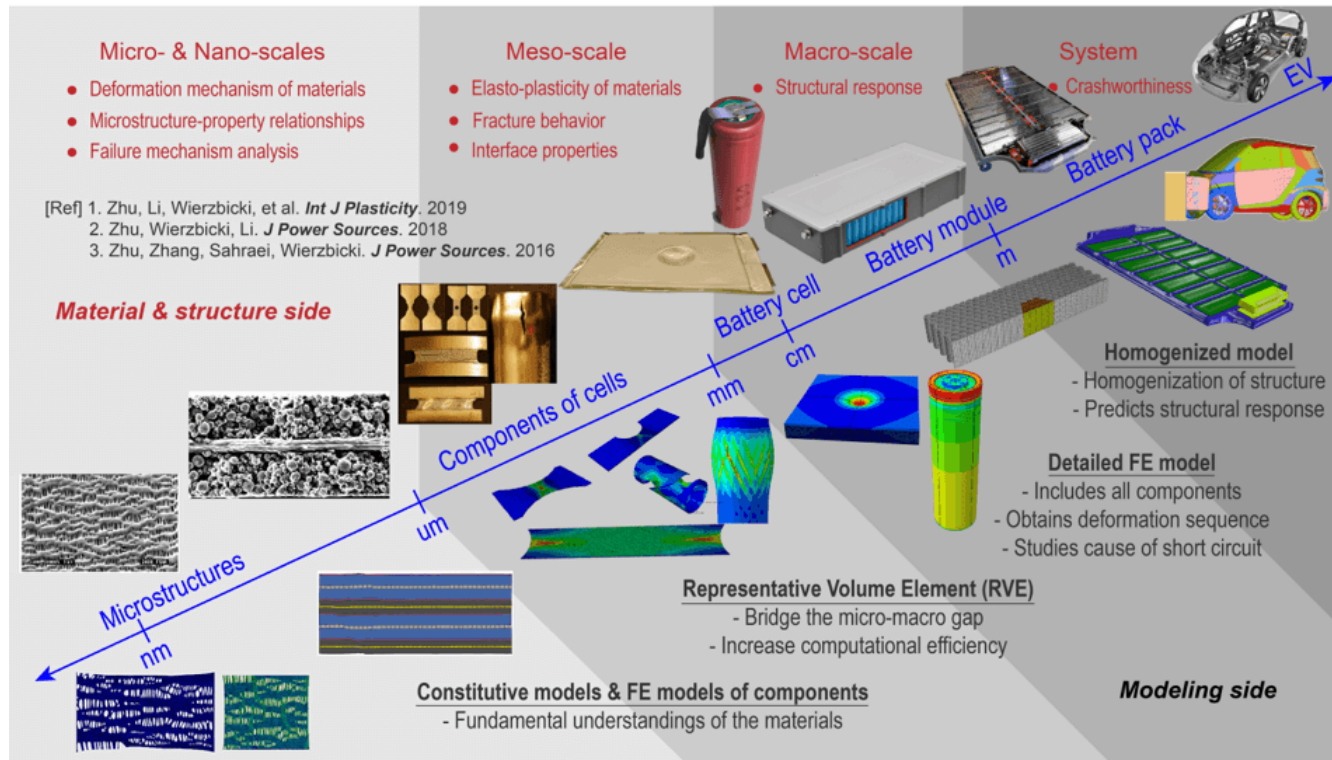
The Scale of the Model Should Match the Question



- Continuum models describe large-scale deformation and transport.
- Atomistic models describe individual atoms, bonds, and local environments.
- Outputs from one scale can become parameters or insight for another.

A molecular simulation is not a replacement for every other model; it is a model at a particular scale.

The Same Idea Appears in Modern Technologies



Example: batteries

- Device performance depends on molecular and materials processes.
- Ion motion, interfacial structure, and mechanical response all matter.
- Atomistic simulations help isolate molecular-level mechanisms.

The course will focus on the atomistic and molecular parts of this hierarchy.

Case Study: Stiffness as a Bridge Between Scales

Engineering question

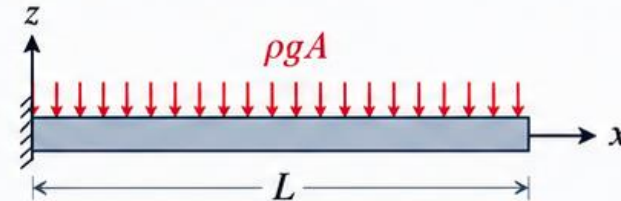
How much will a beam deform under a known load?

The continuum equation can predict the displacement field, but it requires the material parameter E.

The unknown parameter is Young's modulus.

E relates how force and deformation are related

Beam deformation problem – continuum model

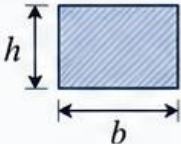


Question: Displacement field

Governing equation (PDE)

$$-EI_{zz} \frac{\partial^4 u_z}{\partial x^4} + q_z = 0$$

Geometry

$$I_{zz} = \frac{bh^3}{12}$$


BC - load

$$\rho g A$$

Integration & BCs

Displacement field



Analytical solution

$$u_z(x) = -\frac{\rho g A}{24EI_{zz}} x^4$$

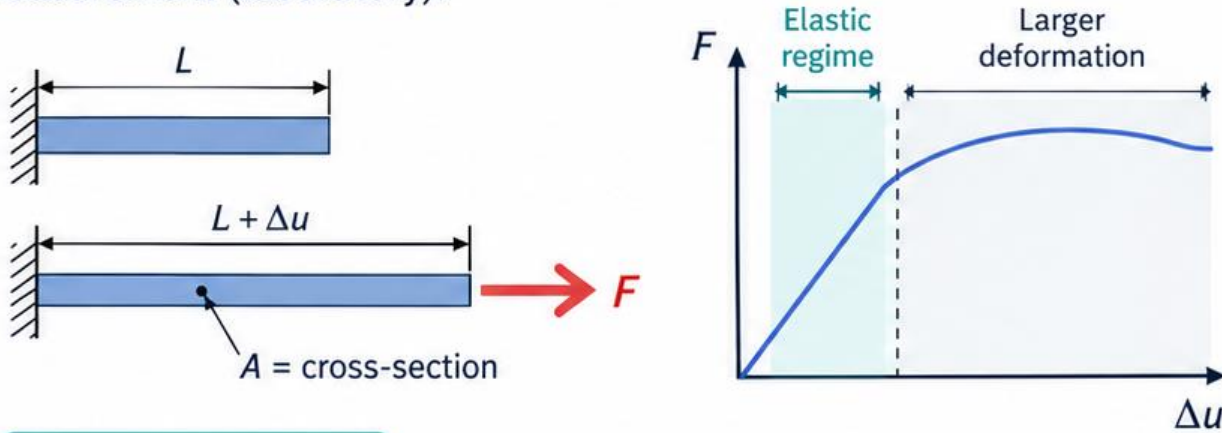
$E = \text{unknown parameter}$

Where Does Young's Modulus Come From?

Classical route

How to determine Young's modulus E ?

Measurement (laboratory):



$$F = k_s \Delta u$$

$$E = \frac{L k_s}{A}$$

Young's modulus E

(~ stiffness = proportionality between force and displacement)

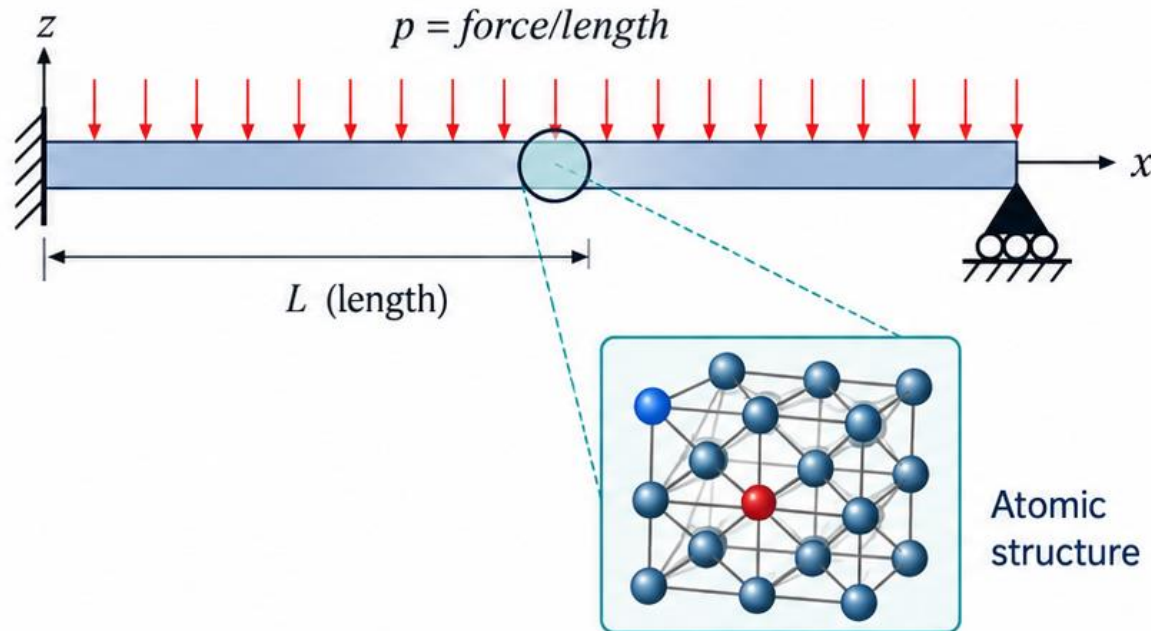
- Apply a known force or strain.
- Measure deformation.
- Extract a slope in the elastic regime.
- Use the measured E in a continuum model.

But what if the material is difficult, expensive, unsafe, or not yet synthesized?

Alternative Route: Atomistic Simulations

How to determine E ? – alternative approach

Atomistic simulation – *new engineering paradigm*



Idea: consider the behavior of a collection of atoms inside the beam as deformation proceeds.

Atomistic simulation replaces the continuum material by a collection of interacting particles.

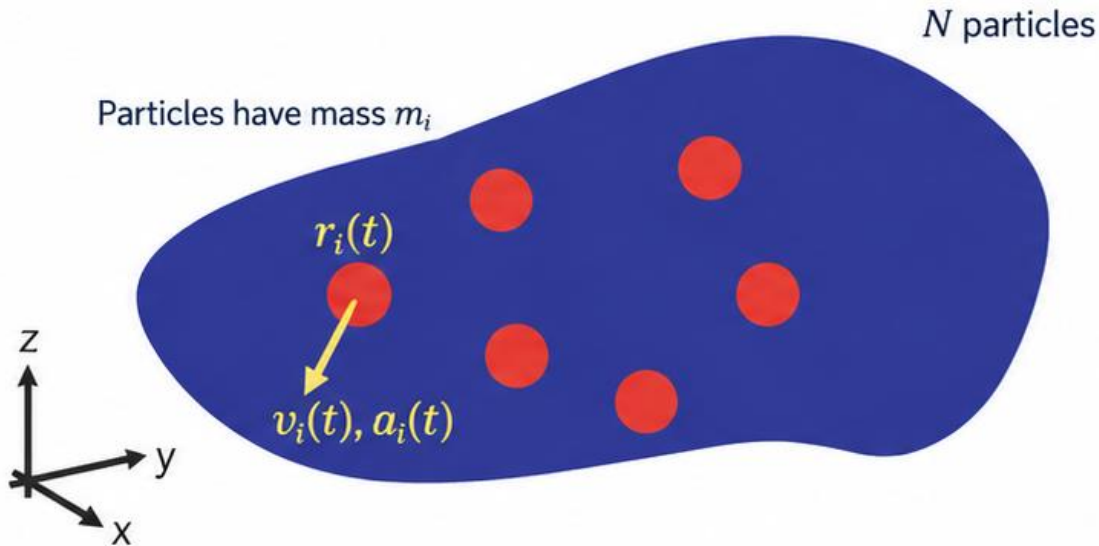
- Atoms have positions.
- Interactions determine forces.
- Forces determine motion or mechanical response.
- Macroscopic properties can be estimated from microscopic behavior.

The atomistic model makes the material parameter computable from molecular structure.

Molecular Dynamics

Molecular dynamics simulation

- Newton's laws: $F = ma$.
- Chemistry: atomic interactions determine the forces; in practice we compute F from an energy model.



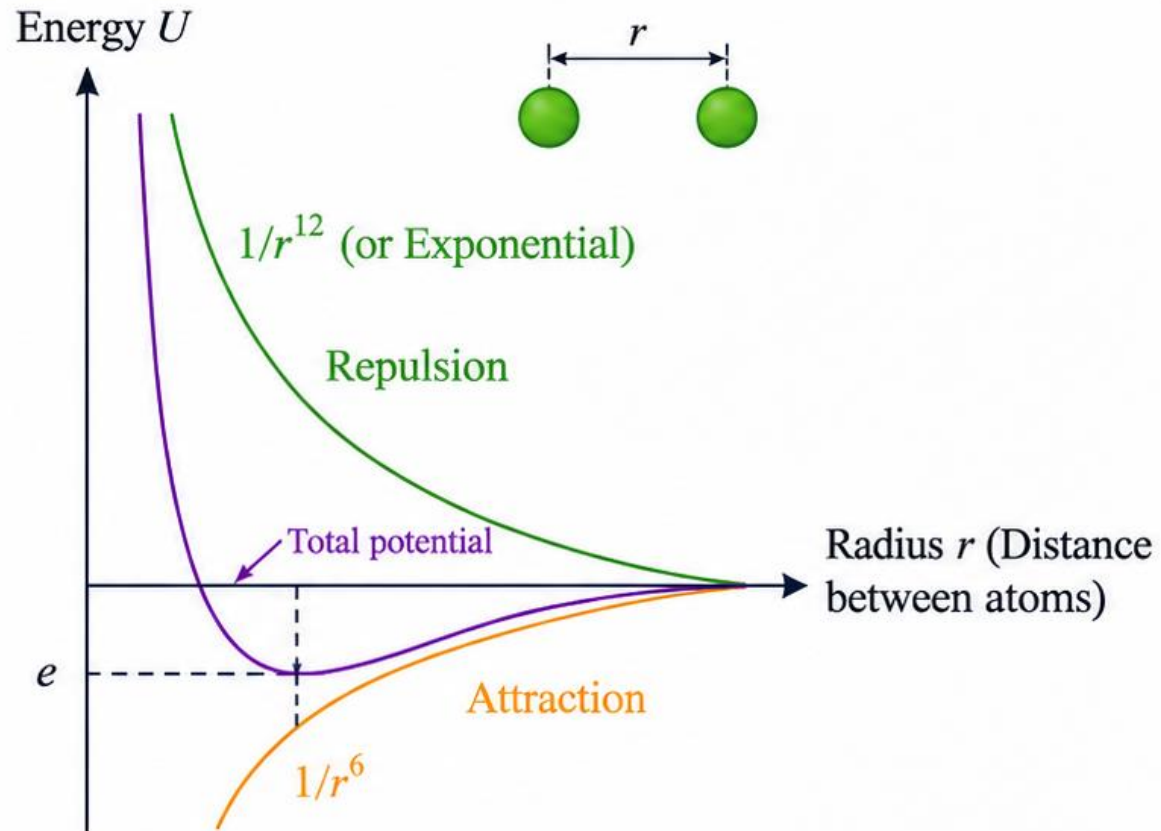
MD asks how atoms move when forces act on them.

$$\mathbf{F} = m\mathbf{a}$$

- Define a potential-energy model.
- Compute forces from energy gradients.
- Integrate equations of motion in time.
- Analyze trajectories to estimate properties.

At the Center is a Potential Energy Model

Pair potential energy as a function of separation

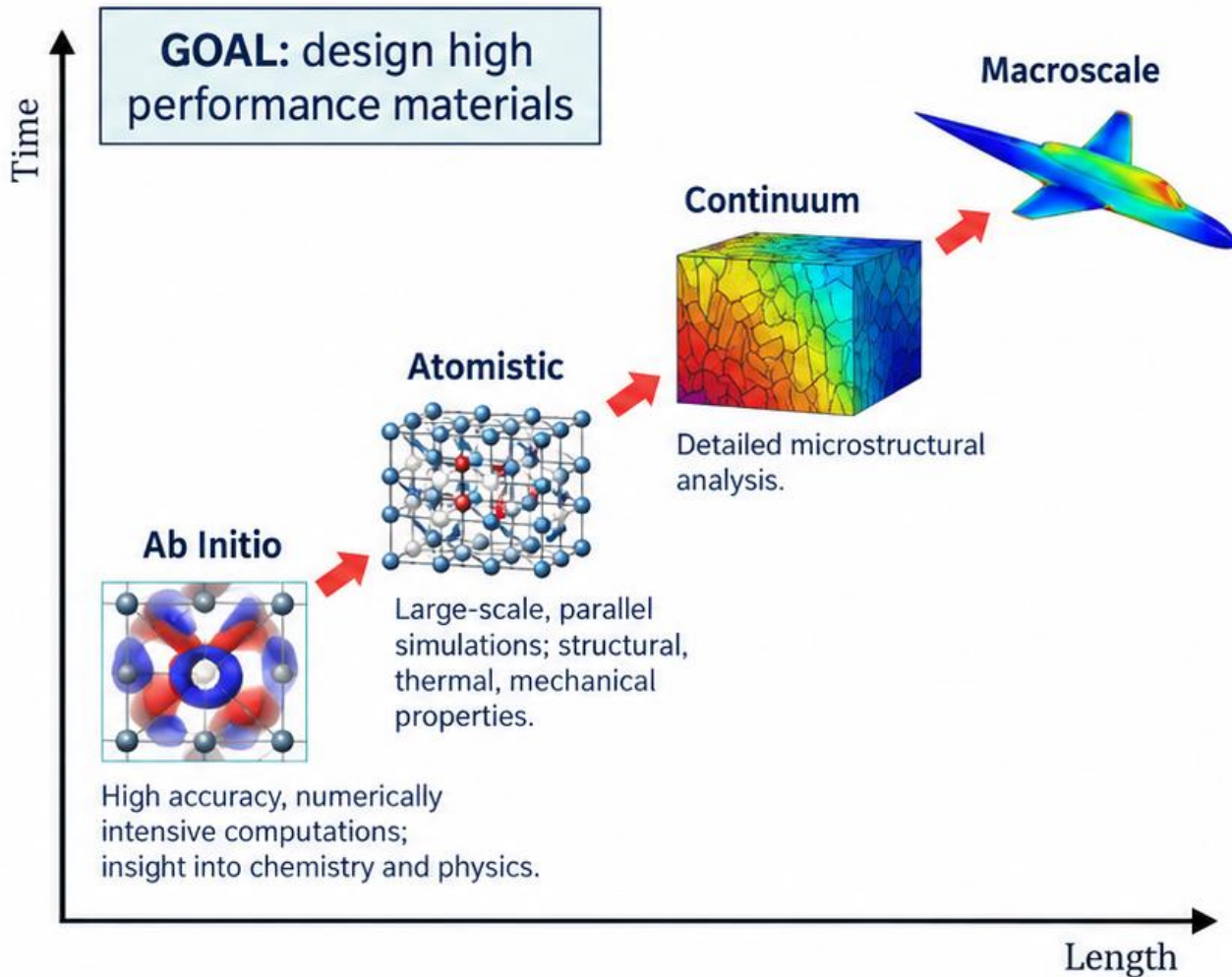


The potential tells us what configurations are energetically favorable or unfavorable.

- Repulsion prevents atoms from overlapping.
- Attraction stabilizes favorable separations.
- Forces are obtained from the slope of the energy landscape.
- The chosen model determines what physics can be studied.

Energy model $\rightarrow$ forces $\rightarrow$ motion $\rightarrow$ observables

Where do Interaction Models Come From?



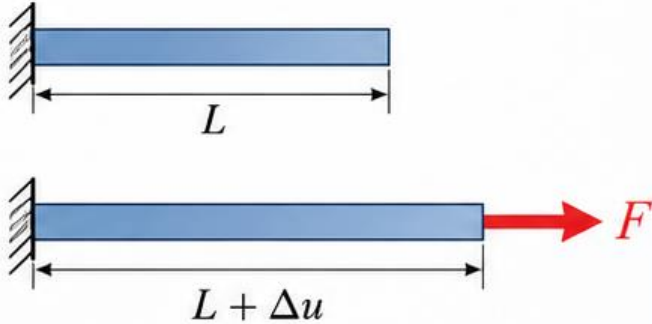
In principle, bonding is electronic. In this course, we usually work one level higher.

- **Electronic structure**
Provides fundamental information about bonding and energies.
- **Classical molecular simulation**
Uses simplified potential-energy models to simulate larger systems and longer times.

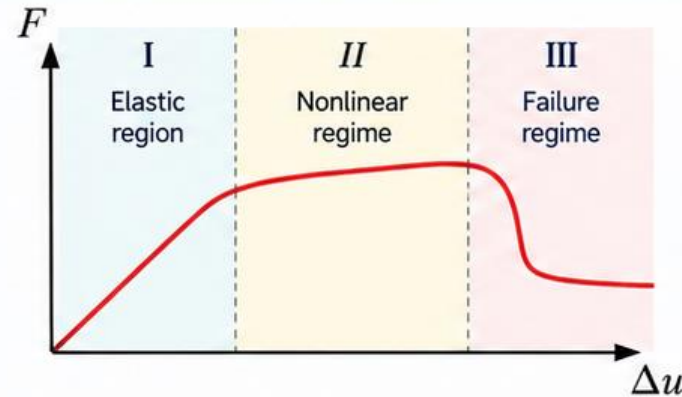
Different levels of description carry different costs and approximations.

Example: Stretching a Nanowire

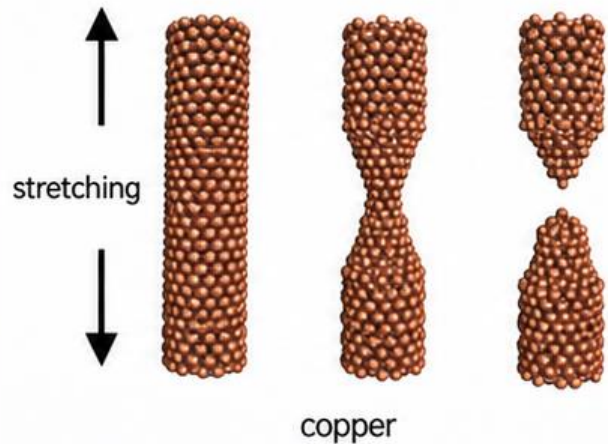
Rod stretching



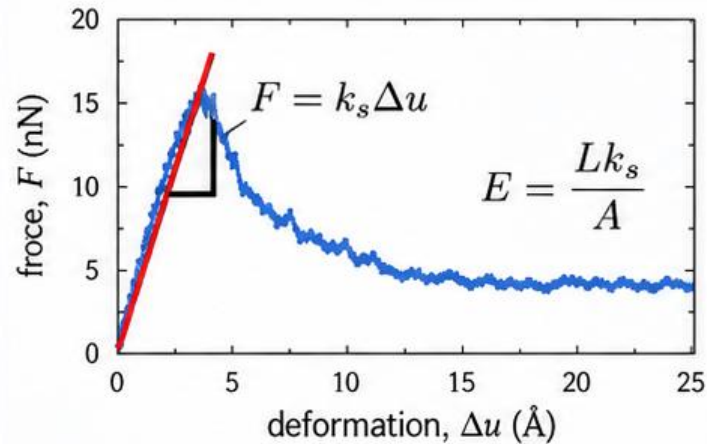
Schematic force–deformation response



Atomistic simulation snapshots (copper)



Force–deformation curve from simulation



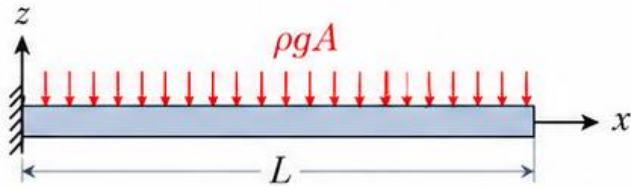
A simulation can generate a force–deformation response.

- Elastic region → estimate stiffness.
- Nonlinear regime → deformation mechanisms.
- Failure regime → bond breaking and structural rearrangement.

The output is not just a movie; it is quantitative data.

Multiscale Dimension

Recap: beam deformation problem – continuum model

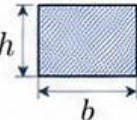


Question: Displacement field $u_z(x)$

Governing equation (PDE)

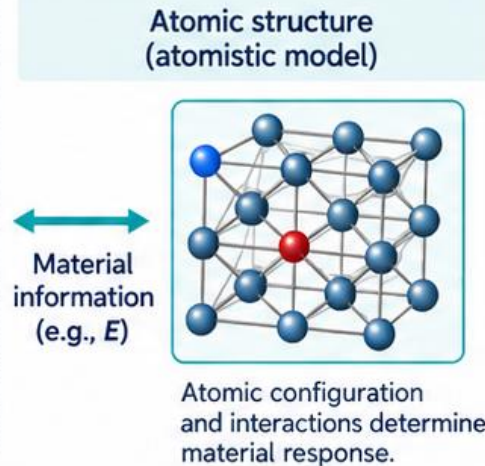
$$-EI_{zz} \frac{\partial^4 u_z}{\partial x^4} + q_z = 0$$

Geometry

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Displacement field



Analytical solution

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E = unknown parameter

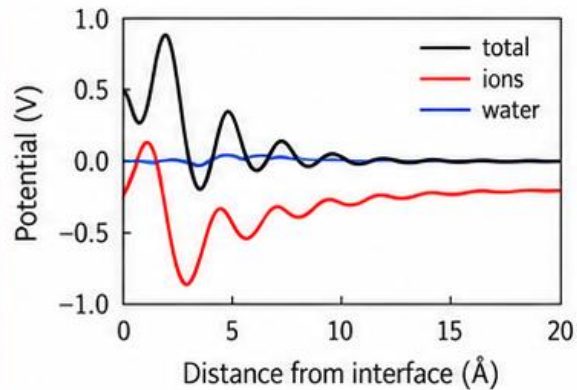
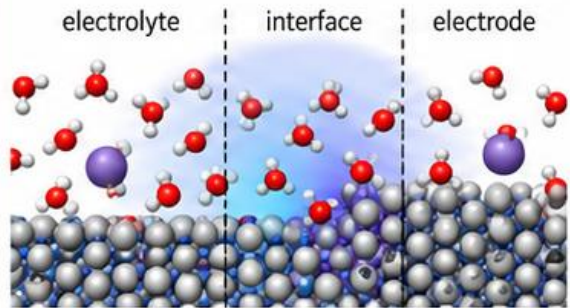
Atomistic simulation supplies material information that can feed larger-scale engineering models.

- This is not merely visualization.
- The goal is to calculate a property or mechanism.
- The result is meaningful only within the assumptions of the model.

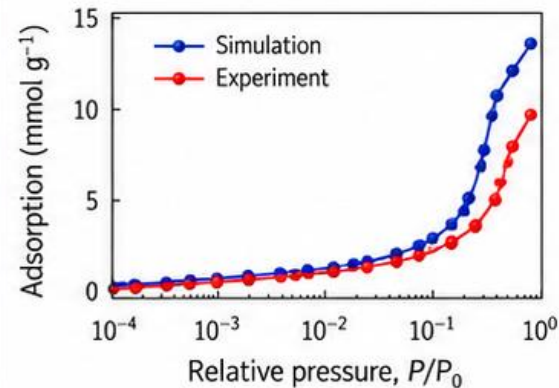
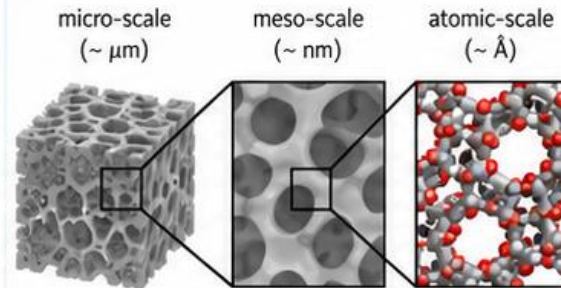
Atomistic model → property/mechanism → continuum or engineering interpretation

Representative Applications

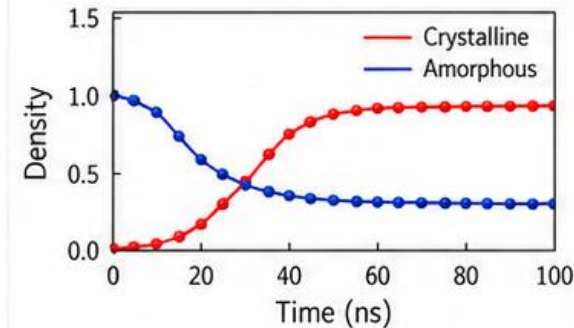
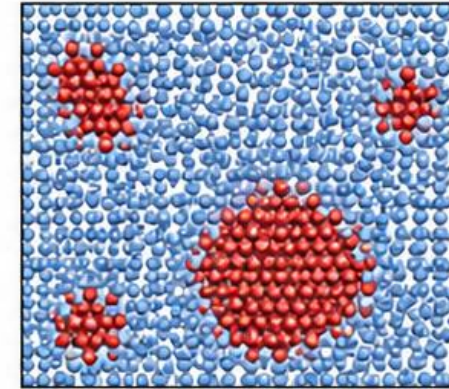
Energy materials and interfaces



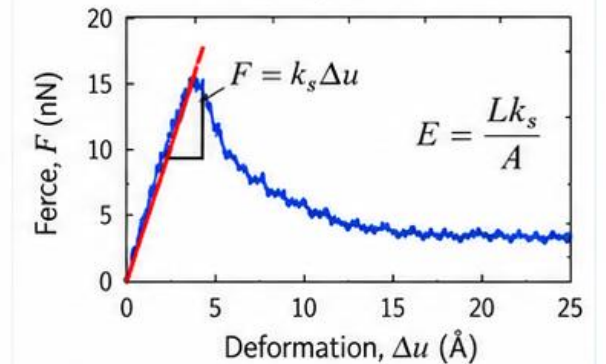
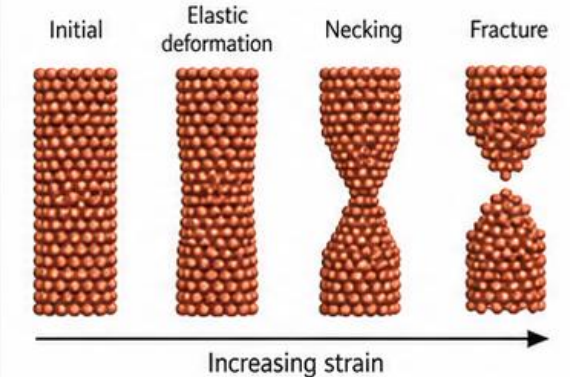
Porous and nanostructured materials



Crystallization and phase behavior



Failure and mechanical response

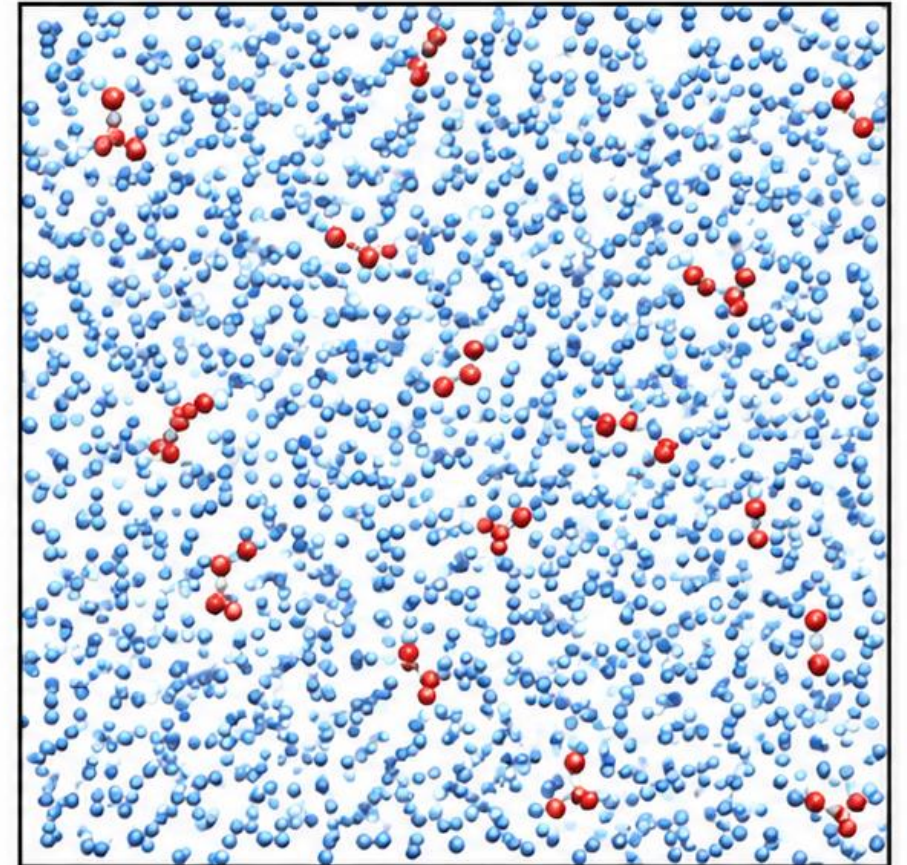


The common thread is not the material type; it is the connection between microscopic structure, dynamics, and observable behavior.

What This Course Emphasizes

Not just “press run”
but knowing what the simulation means.

- What model am I using?
- What quantity am I estimating?
- Is the result converged and reproducible?
- Did my algorithm change the property I want to measure?
- Do I need equilibrium sampling, free energies, rates, or ML-assisted analysis?



Takeaway Messages

- A model simplifies reality while retaining the features needed for the question.
- A simulation is the numerical experiment performed on that model.
- The interaction potential already encodes physical assumptions before any trajectory is generated.
- Equilibrium separation, force, and local curvature connect microscopic interactions to stiffness.
- Method choice in this course will always be tied to the scientific question.

Key bridge from today: potential $\rightarrow$ force $\rightarrow$ stiffness $\rightarrow$ macroscopic response.

Next week: statistical mechanics and Monte Carlo sampling.

Hands-on Session

From interatomic interactions to stiffness

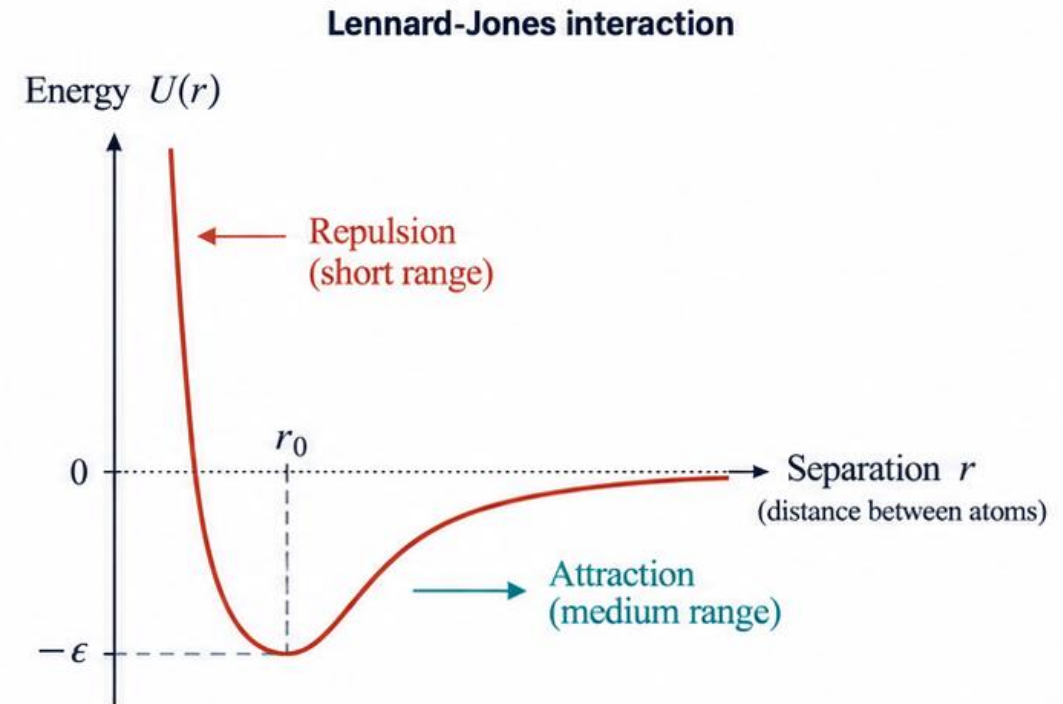
We will not run full molecular dynamics yet. We start with the simplest microscopic model: an interaction potential and the force derived from it.

$$U(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

$$F(r) = -dU/dr$$

- Plot the interaction energy and the corresponding force.
- Identify the equilibrium separation r_0 .
- Relate local curvature to pair stiffness.
- Build a simple 1D chain and obtain a force–extension response.

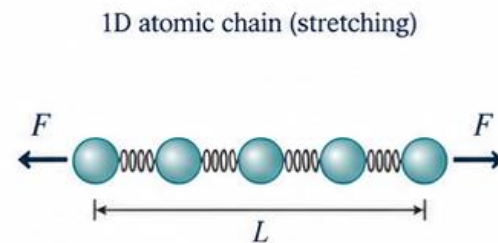
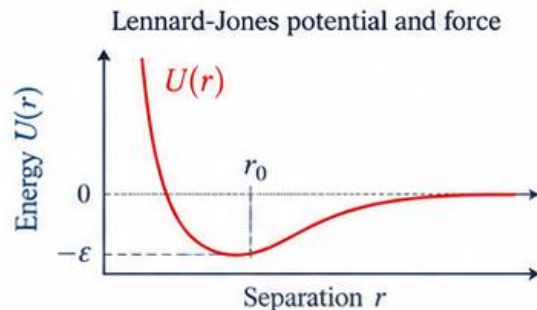
Before running MD, the energy model already encodes physical assumptions.



HW1 Preview

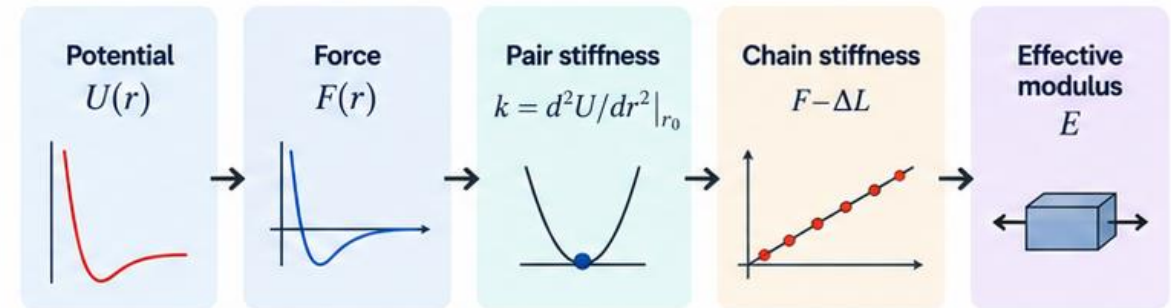
In today's notebook

- **Implement** the Lennard-Jones energy and force.
- **Identify** the equilibrium separation r_0 and interpret local curvature.
- **Estimate** pair stiffness from the second derivative of $U(r)$.
- **Stretch** a 1D atomic chain and compute a force–extension response.
- **Connect** microscopic stiffness to the Young's modulus example.



HW1 will extend this

- **Compare** multiple parameter sets (ϵ, σ).
- **Test** the harmonic approximation against the full potential.
- **Study** how chain stiffness changes with system size N .
- **Define** an effective modulus and distinguish it from structural stiffness.
- **Provide** short physical interpretations, not just code.



HW1 asks you to complete and extend this notebook, generate figures, and explain the physics.

The goal is not coding complexity; the goal is physical interpretation.